Ground Water Quality Alert

Central Ground Water Board (CGWB) • Government of India

Location	All India	Analysis Period	From: 15/10/2026 To: 29/10/2026
Issue Date	29/10/2026	Issuing Office	CGWB, Regional Office-All India

Groundwater Quality Analysis Report Summary

A comprehensive analysis of 801 groundwater samples was conducted across various states in CGWB Regional Chemical Laboratories to assess water quality as per the BIS Drinking Water Standard (IS 10500:2012). Analysed parameters included pH, Electrical Conductivity, Total Hardness, Calcium, Magnesium, Sodium, Potassium, Carbonate, Bicarbonate, Sulphate, Chloride, Nitrate, Fluoride and Trace/Heavy Metals such as Arsenic, Uranium and Iron. Additionally, specific testing for Heavy/Trace Metals was performed on groundwater samples from Delhi, Gujarat, Haryana, and Rajasthan.

Key Compliance Findings:

Out of the **801** total samples analyzed, **780** were found to be contaminated. These results are categorized as follows:

- Exceeding Acceptable Limit (but up to permissible limit): **259** samples
- Exceeding Permissible Limit: **521** samples

The following table provides a breakdown of contaminant detection by State/Union Territory. Please refer to the attached Excel sheet for the comprehensive, parameter-wise analytical data.

Para I

Sl. No.	Name of State/UT	Total samples analysed	No. of samples contaminated			Parameters of contamination
			Above Acceptable	Above Permissible	Total Contaminated	
1	Delhi	40	15	21	36	Arsenic, Calcium, Electrical conductivity, Iron, Magnesium, Manganese, Nitrate, Sulphate, Total Hardness, Uranium, pH, Chloride, Fluoride
2	Gujarat	17	5	11	16	Calcium, Electrical conductivity, Fluoride, Magnesium, Nitrate, Total Hardness, Chloride, Manganese
3	Haryana	99	28	65	93	Arsenic, Calcium, Chloride, Electrical conductivity, Fluoride, Iron,

Sl. No.	Name of State/UT	Total samples analysed	No. of samples contaminated			Parameters of contamination
			Above Acceptable	Above Permissible	Total Contaminated	
						Magnesium, Manganese, Nitrate, Sulphate, Total Hardness, Uranium, pH, Zinc
4	Rajasthan	645	211	424	635	Arsenic, Calcium, Chloride, Electrical conductivity, Fluoride, Iron, Magnesium, Manganese, Nitrate, Sulphate, Total Hardness, Uranium, pH, Copper, Zinc
Total		**801**	**259**	**521**	**780**	

Para II

The anomalies may be termed major in some case.

Risk Perception and Remedial Measures

Risk Perception & Remedial Measures for pH

Remedial Measures:

Treatment of acidic water: Passing through Neutralizing Filter containing Calcite or Magnesia, or adding Soda Ash. Treatment of basic water: Installation of an acid injector (e.g., white vinegar or citric acid).

Risk Perception & Remedial Measures for EC

Remedial Measures:

Since Electrical Conductivity is directly related to dissolved ions (TDS), methods like Reverse Osmosis (RO) and Deionization (Ion Exchange) effectively lower it.

Risk Perception & Remedial Measures for Chloride

Risk Perception: High concentration of chloride concentration in draining water.
Remedial Measures:

Reverse Osmosis (RO), Distillation, and specific Ion Exchange (anion exchange) resins are effective for chloride removal. Standard boiling does not remove chlorides.

Risk Perception & Remedial Measures for Fluoride

Risk Perception: Dental and Skeletal Fluorosis: Studies suggest high levels of fluoride may lead to discoloration and damage to teeth and affect bone health.
Remedial Measures:

Defluoridation techniques can be adopted: Adsorption technique, Ion-exchange technique, Precipitation technique and other techniques, which include electro chemical defluoridation and Reverse Osmosis.

Risk Perception & Remedial Measures for Nitrate

Risk Perception: High nitrate causes cyanosis (methaemoglobinaemia) or Blue baby disease. It effects digestive system. Development of cancer, especially digestive cancers, such as colorectal stomach and esophagus cancer, complications during pregnancy.
Remedial Measures:

Reverse osmosis systems for drinking water to be used. Biological denitrification and ion-exchange can also be applied.

Risk Perception & Remedial Measures for Sulphate

Risk Perception: High concentration of sulphate concentration in draining water may be more sensitive to the cathartic effects. Children, transients and the elderly are high risk of dehydration from diarrhoea may be caused by high concentration of sulfate in drinking-water.
Remedial Measures:

The techniques may be adopted: Ion-exchange technique, Nano-filtration technique, Reverse Osmosis and electro-dialysis may be used.

Risk Perception & Remedial Measures for Calcium

Remedial Measures:

Water softening using cation exchange resins, Reverse Osmosis (RO), or chemical lime softening.

Risk Perception & Remedial Measures for Magnesium

Remedial Measures:

Water softening using cation exchange resins, Reverse Osmosis (RO), or chemical lime softening.

Risk Perception & Remedial Measures for Total Hardness

Risk Perception: Reduced lathering of soaps, buildup of scale on heating elements and boilers, reduced water flow in distribution pipes due to scale buildup. High levels of sodium in drinking water may harm your health like Cathartic and diuretic effect.
Remedial Measures:

Hardness minerals can be removed with a water softener and dilution technique and Reverse Osmosis., which replaces the calcium and magnesium (and iron, manganese, radium and other positive ions) with sodium.

Risk Perception & Remedial Measures for Copper

Remedial Measures:

Reverse Osmosis (RO), Cation Exchange (water softeners), and raising the pH of the water (to prevent pipe corrosion which is a common source of copper).

Risk Perception & Remedial Measures for Zinc

Remedial Measures:

Ion Exchange (water softeners), Reverse Osmosis (RO), and Distillation are highly effective at removing zinc from drinking water.

Risk Perception & Remedial Measures for Arsenic

Risk Perception: Long-term exposure of Arsenic is associated with skin, bladder, lung, risk of heart disease and hypertension and other cancers.
Remedial Measures:

The techniques may be adopted: Adsorption technique, Ion-exchange technique, Reverse Osmosis and using alternative sources such as rainwater harvesting or deep aquifer water.

Risk Perception & Remedial Measures for Iron

Risk Perception: An excessive amount of exposure to iron can increase the risk of Parkinson's, cardiovascular disease, diabetes mellitus, pigmentation changes, Alzheimer's, as well as renal, liver, respiratory, and neurological issues.
Remedial Measures:

Oxygenation of groundwater followed by precipitation and Reverse osmosis systems.

Risk Perception & Remedial Measures for Manganese

Risk Perception: Continue use can damage liver, lungs and kidneys, effects the metabolism of iron for formation of hemoglobin and leads to neurological conditions- impaired memory, impaired motor coordination and delayed reaction time- results in disease called Manganism.
Remedial Measures:

Treating the water at its source, filtration system and water treatment method.

Risk Perception & Remedial Measures for Uranium

Risk Perception: The kidney and lungs are found to be the major target organs of Uranium toxicity and it may cause cancer from radioactive uranium. In animal bones, Uranium influences inhibition of osteoblastic activity, thereby reducing bone volume.
Remedial Measures:

The technologies for uranium removal: Coagulation/filtration, Lime softening, Anion exchange, Reverse osmosis and Activated alumina may remove 80-90% uranium from water.